Exploratory Data Analysis & Business Intelligence on Retail Sales Data

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Internship Task 2

1. Objective:

The objective of this project is to perform Exploratory Data Analysis (EDA) on a retail sales dataset to identify patterns, trends, and business insights using Python and SQL. The project also aims to present the findings through visualizations and a dashboard.

2. Dataset Description :

The Superstore dataset contains information about customer orders, sales, profit, discount, quantity, product categories, and regions. It is used to analyze business performance and customer purchasing behavior. It contains 5 rows and 21 columns. Each row contains important columns such as Sales, Profit, Discount, Category, Region, and Quantity.

3. Descriptive Statistics :

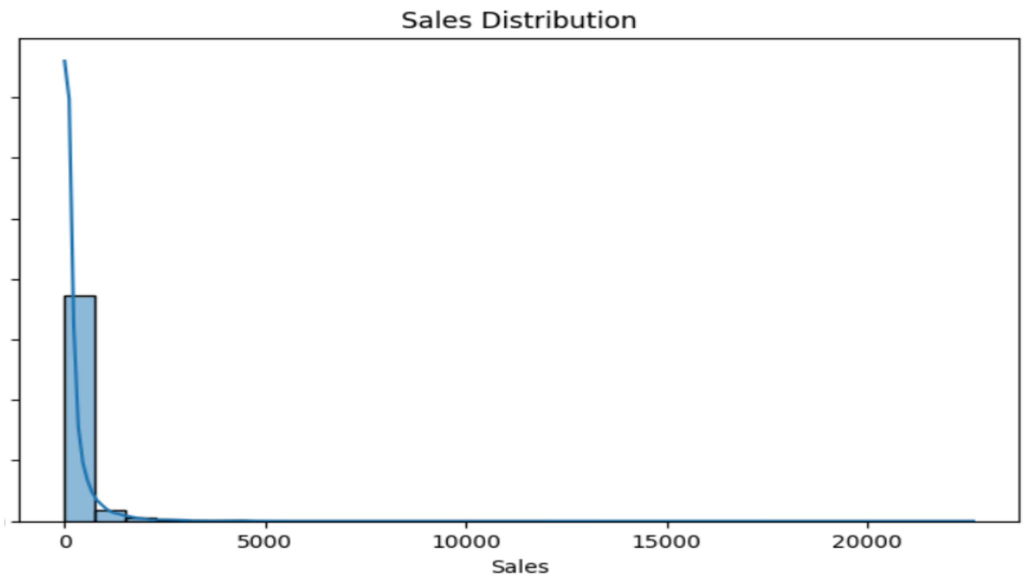
Descriptive statistics were used to summarize the numerical features of the dataset. Metrics such as mean, standard deviation, minimum, and maximum values were analyzed.

```
print(df.describe())
```

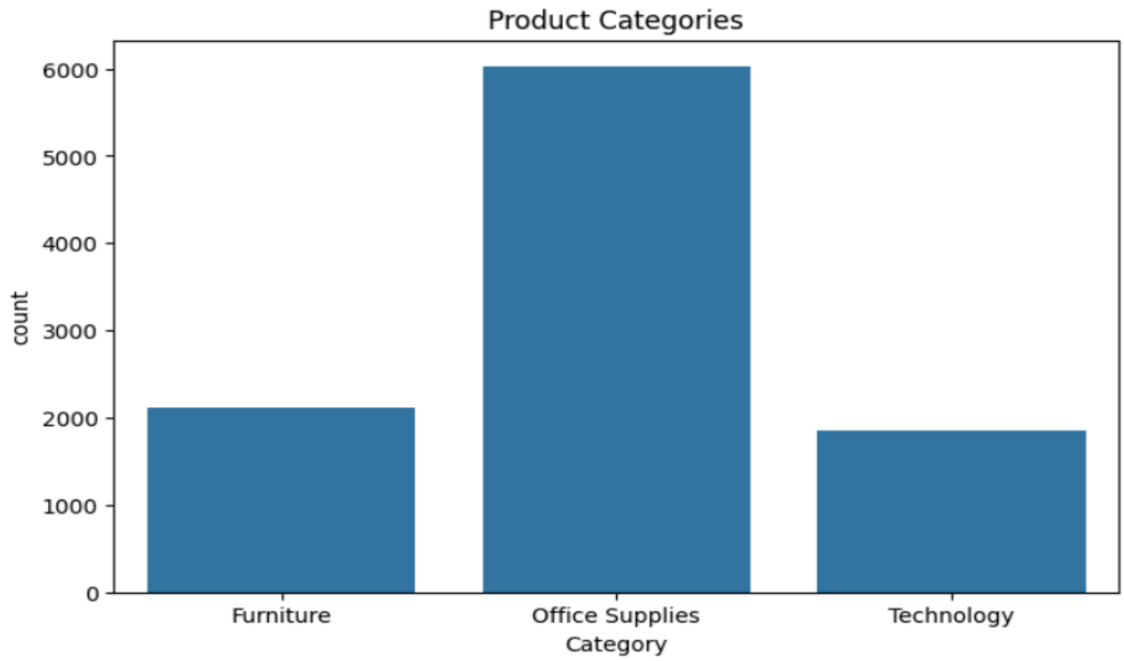
```
            Row ID   Postal Code         Sales     Quantity     Discount  \
count  9994.000000   9994.000000   9994.000000  9994.000000  9994.000000
mean   4997.500000  55190.379428    229.858001     3.789574     0.156203
std    2885.163629  32063.693350    623.245101     2.225110     0.206452
min       1.000000   1040.000000      0.444000     1.000000     0.000000
25%    2499.250000  23223.000000     17.280000     2.000000     0.000000
50%    4997.500000  56430.500000     54.490000     3.000000     0.200000
75%    7495.750000  90008.000000    209.940000     5.000000     0.200000
max    9994.000000  99301.000000  22638.480000    14.000000     0.800000

            Profit
count  9994.000000
mean     28.656896
std     234.260108
min   -6599.978000
25%       1.728750
50%       8.666500
75%      29.364000
max    8399.976000
```

4. Univariate Analysis :

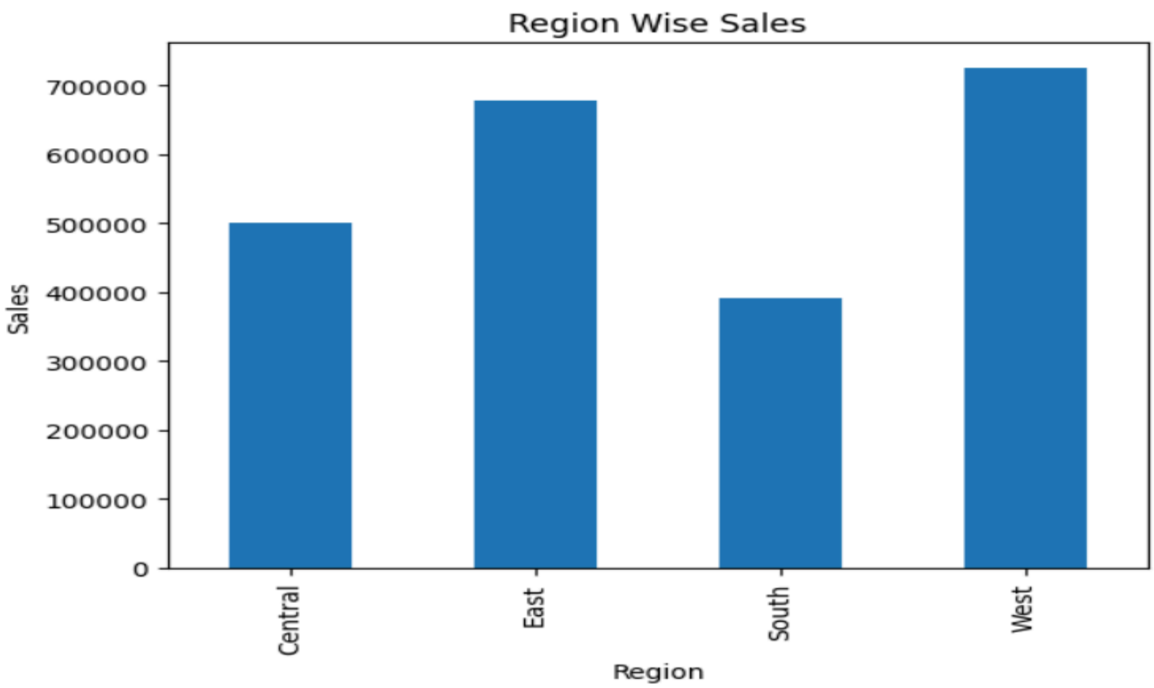
- Sales Distribution Histogram



- Category Count Plot



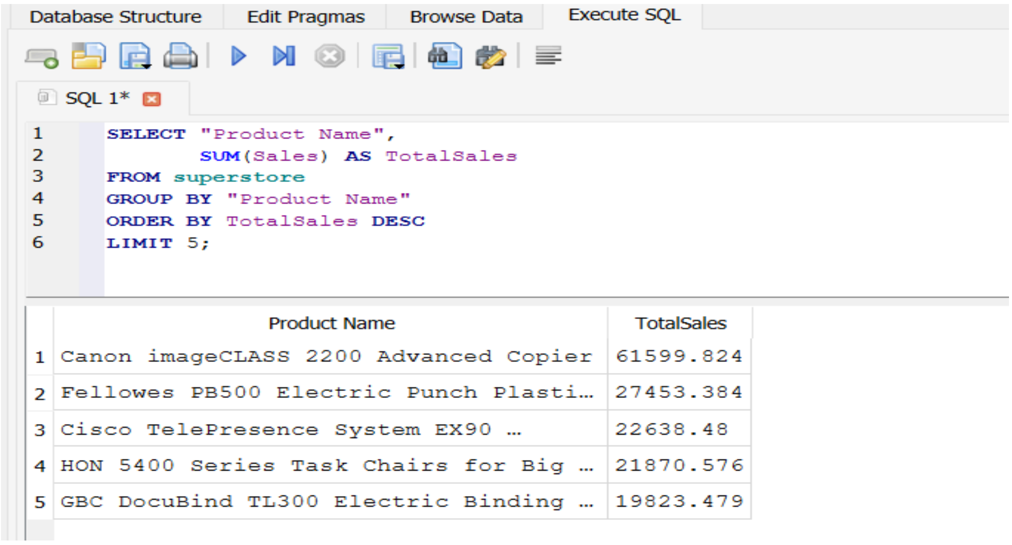
- Region-wise Sales Chart



5. SQL Business Analysis :

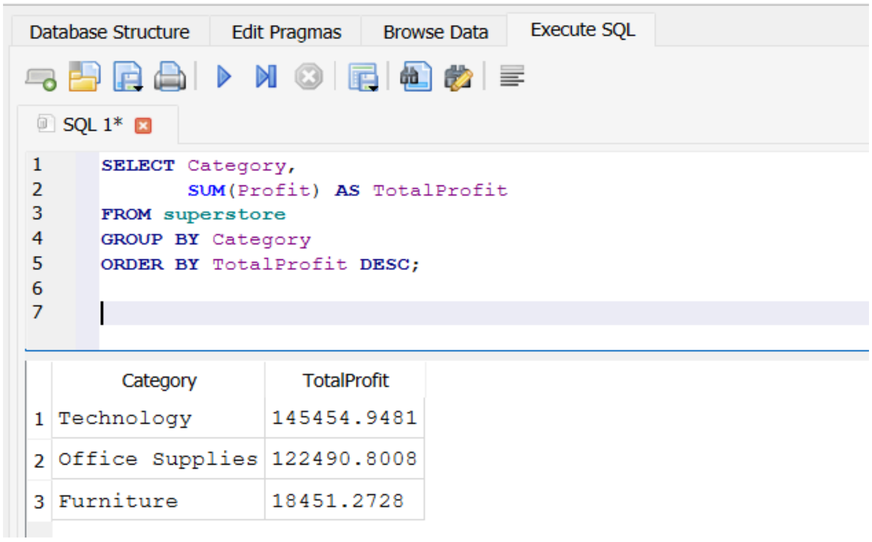
- Business Question 1: Which are the Top 5 Products by Sales?

```
SELECT "Product Name",
       SUM(Sales) AS TotalSales
FROM superstore
GROUP BY "Product Name"
ORDER BY TotalSales DESC
LIMIT 5;
```



- Business Question 2: Which Category Earned the Highest Profit?

```
SELECT Category,
       SUM(Profit) AS TotalProfit
FROM superstore
GROUP BY Category
ORDER BY TotalProfit DESC;
```



- Business Question 3: Which Region Generates the Highest Sales?

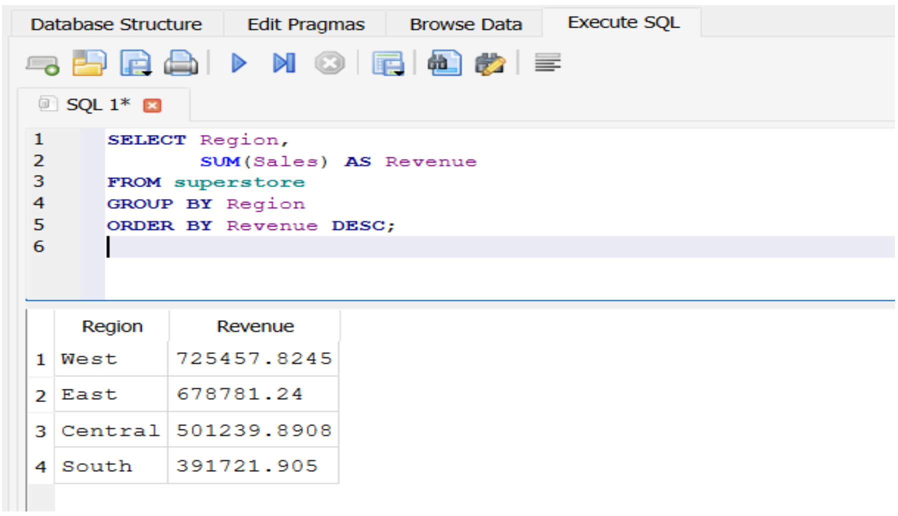
SELECT Region,

SUM(Sales) AS Revenue

FROM superstore

GROUP BY Region

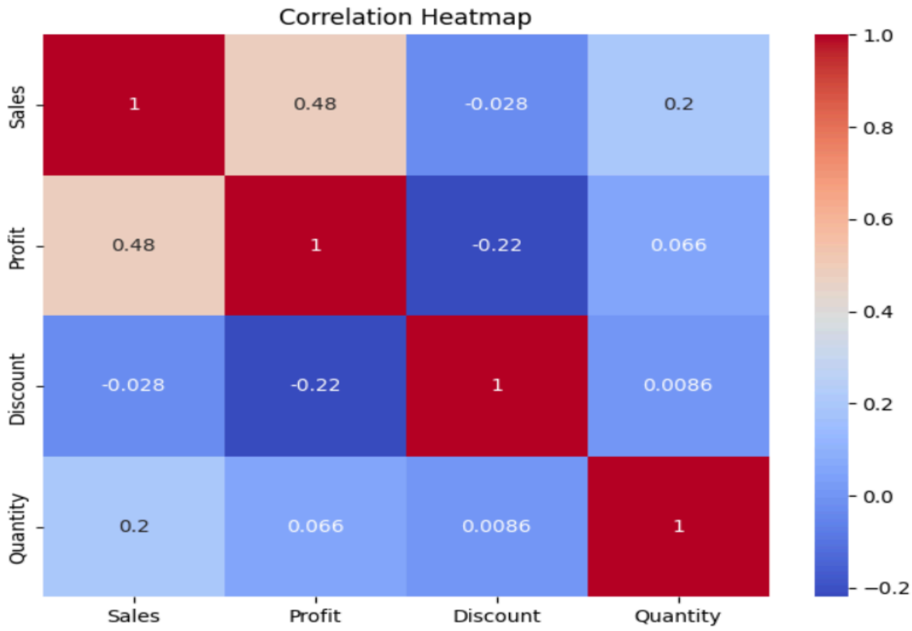
ORDER BY Revenue DESC;



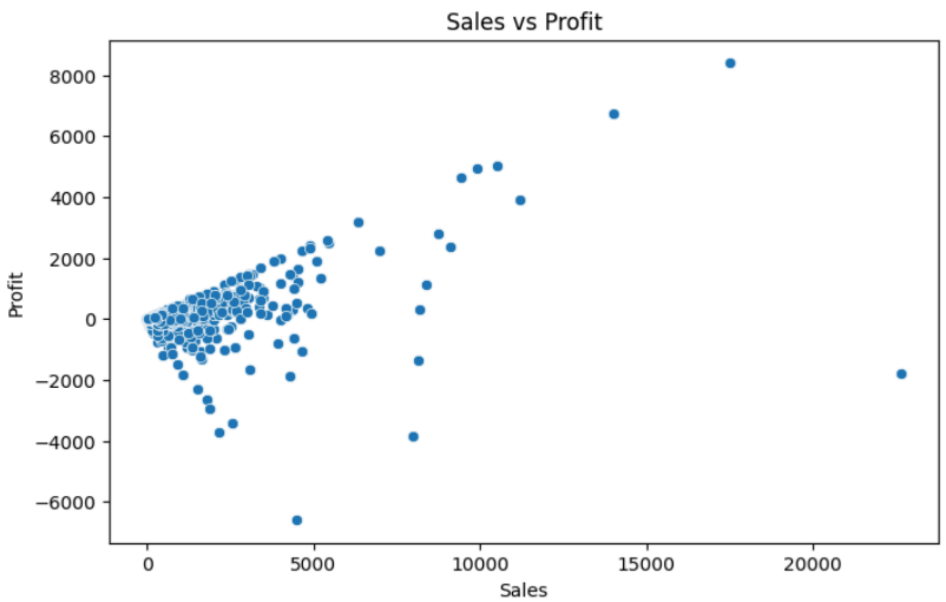
6. Correlation & Multivariate Analysis :

Correlation analysis showed a positive relationship between Sales and Profit. Discount showed a negative relationship with Profit, indicating that excessive discounts can reduce profitability.

- Correlation Heatmap

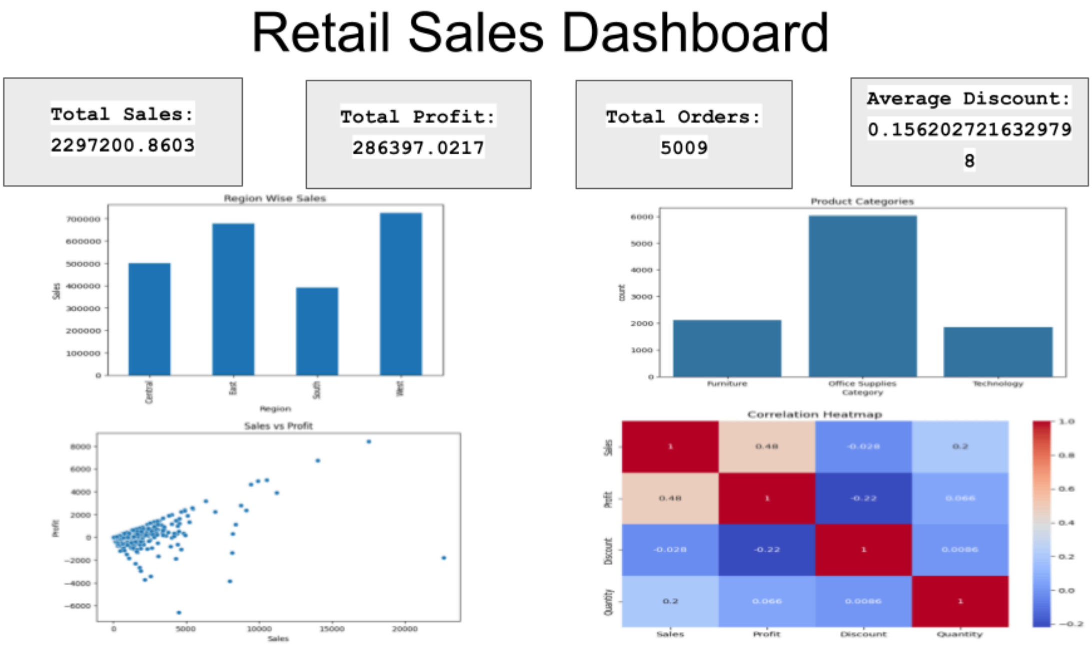


- Sales vs Profit Scatter Plot



7. Dashboard :

A static dashboard was created to summarize key performance indicators and important visualizations.



8. Key Insights :

- Technology category generated the highest profit.
- West region contributed the highest sales.
- Sales and Profit showed a positive correlation.
- Higher discounts often reduced profitability.
- Office Supplies had the highest number of transactions.
- A small number of products generated a large share of revenue.

9. Conclusion :

This project successfully analyzed retail sales data using EDA and SQL techniques. The findings revealed important business trends and provided actionable insights that can support decision-making.